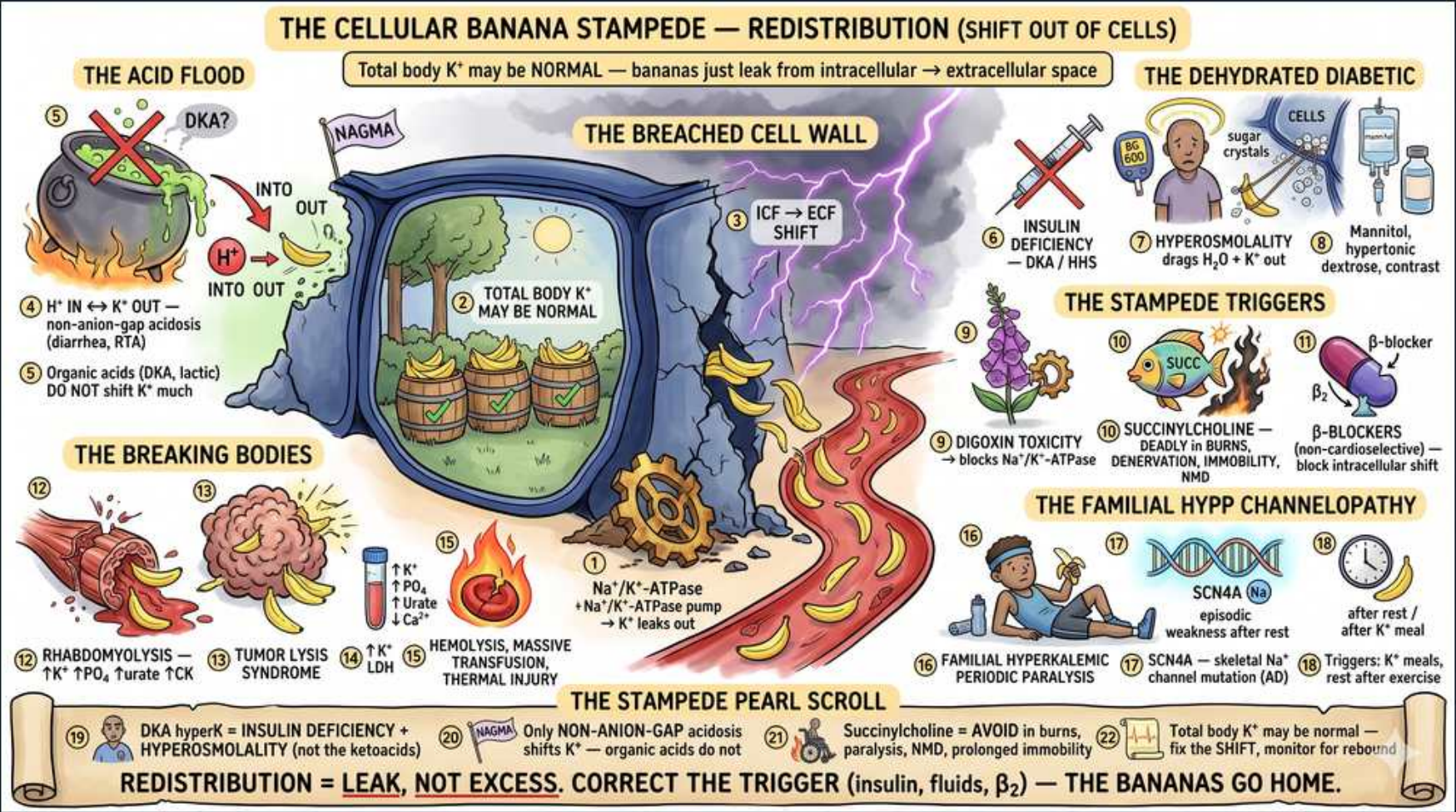


Hyperkalemia — redistribution (transcellular shift)



1. Total body K+ normal

WHAT YOU SEE

Banner: total body K+ may be normal

WHAT IT ENCODES

In redistribution (transcellular shift), K+ moves from the intracellular compartment (where ~98% of body K+ lives, ICF ~140 mEq/L) into the extracellular space, so the measured serum K+ rises even though total body K+ is unchanged. This is the key conceptual split from true K+ excess (renal/intake causes). Recognizing a shift changes management: you fix the trigger and watch for rebound HYPOkalemia once cells take the K+ back up.

BOARD TRAP

WRONG answer: aggressively dumping total-body potassium with binders or dialysis for a pure shift. The discriminator is that the deficit is positional, not absolute — over-removal plus reversal of the trigger (e.g., giving insulin in DKA) causes dangerous rebound hypokalemia. Stabilize/shift and treat the cause first.

2. Acidosis H+/K+

WHAT YOU SEE

Acid flood, H+ in / K+ out

WHAT IT ENCODES

In MINERAL (non-anion-gap) acidosis, excess extracellular H+ enters cells to be buffered, and to maintain electroneutrality K+ is exchanged outward — roughly a 0.6 mEq/L rise in K+ per 0.1 fall in pH (highly variable). The classic exam pairing is hyperchloremic NAGMA from diarrhea or RTA. The mnemonic: H+ IN, K+ OUT.

BOARD TRAP

WRONG answer: assuming every acidosis raises K+ via the H+/K+ exchange. Organic-anion (high anion gap) acidoses — lactic acidosis and DKA — do NOT meaningfully drive this exchange because the accompanying organic anion follows H+ into the cell. Their hyperkalemia comes from other mechanisms (insulin deficiency, hyperosmolality).

3. NAGMA shifts

WHAT YOU SEE

NAGMA label, diarrhea/RTA note

WHAT IT ENCODES

Only non-anion-gap (hyperchloremic) metabolic acidosis — diarrhea, RTA (type 4 especially), TPN, acetazolamide — reliably shifts K+ out of cells via the H+/K+ exchanger. The retained chloride anion cannot easily follow H+ intracellularly, so K+ exits instead. Tie this to the type 4 RTA panel where hyperkalemia and NAGMA coexist.

BOARD TRAP

WRONG answer: attributing DKA or lactic-acidosis hyperkalemia to the acidosis itself. In DKA, the dominant driver is insulin deficiency plus hyperosmolality; the patient is usually total-body K+ DEPLETED, and K+ will plummet once insulin is started. Discriminator: check the anion gap — gap acidosis means look beyond the H+ shift.

4. Insulin deficiency

WHAT YOU SEE

Dehydrated diabetic, ICF->ECF shift

WHAT IT ENCODES

Insulin normally drives K+ into cells by stimulating the Na/K-ATPase (independent of glucose entry). In insulin-deficient states (DKA, HHS), this pump action is lost and K+ exits cells, so the serum K+ can be high or normal despite a large total-body K+ deficit from osmotic diuresis. This is exactly why insulin is a cornerstone of hyperkalemia treatment.

BOARD TRAP

WRONG answer: seeing a 'normal' or high K+ in DKA and concluding K+ stores are fine. The discriminator is that insulin therapy unmasks profound total-body depletion — anticipate hypokalemia, and hold/replace K+ (start K+ if serum K+ < 5.3, hold insulin if K+ < 3.3) per DKA protocols.

5. Hyperosmolality

WHAT YOU SEE

Sugar crystals dragging K+ out

WHAT IT ENCODES

A rise in extracellular osmolality (severe hyperglycemia, mannitol, hypertonic saline/contrast) pulls water out of cells via solvent drag, and K+ is carried along with that water by frictional/convective drag, raising serum K+. This is additive to insulin deficiency in DKA/HHS. Roughly a 0.3–0.6 mEq/L K+ rise per 10 mOsm/kg increase in effective osmolality.

BOARD TRAP

WRONG answer: confusing this with the H+/K+ acidosis mechanism. The discriminator is the trigger — here it is the osmotic gradient (glucose, mannitol), not pH. Correcting hyperglycemia/osmolality alone (plus the insulin it requires) will lower K+ without binders.

6. Na/K-ATPase

WHAT YOU SEE

Pump in the breached cell wall

WHAT IT ENCODES

The Na/K-ATPase actively pumps K+ INTO cells (3 Na+ out, 2 K+ in per ATP) and is the master regulator of internal K+ balance; it is stimulated by insulin and beta-2 agonists. Anything that inhibits it — digoxin, or loss of its activators — causes K+ to accumulate extracellularly. This single pump links insulin deficiency, beta-blockade, and digoxin into one mechanistic story.

BOARD TRAP

WRONG answer: thinking pump inhibition raises K+ only modestly in all cases. Discriminator: massive acute digoxin overdose can block enough pumps to cause life-threatening hyperkalemia, and the K+ level is a recognized prognostic marker of digoxin toxicity severity.

7. Digoxin toxicity

WHAT YOU SEE

Digoxin blocks Na/K-ATPase

WHAT IT ENCODES

Digoxin competitively inhibits the Na/K-ATPase; in acute toxicity this blocks K+ entry into cells and produces hyperkalemia. Serum K+ correlates with mortality in acute digoxin poisoning (K+ > 5.0–5.5 historically predicted high mortality before Fab fragments). The definitive treatment is digoxin-specific antibody (Fab) fragments, which also corrects the hyperkalemia.

BOARD TRAP

WRONG answer: giving IV calcium for the hyperkalemia of digoxin toxicity. Classic teaching warns of the 'stone heart' (calcium worsening digoxin-induced intracellular calcium overload and causing intractable contraction/arrhythmia) — give calcium slowly/cautiously or avoid it, and reach for digoxin-specific Fab instead.

8. Succinylcholine

WHAT YOU SEE

Succinylcholine syringe, denervation

WHAT IT ENCODES

Succinylcholine, a depolarizing neuromuscular blocker, opens acetylcholine receptors and causes K⁺ efflux, normally raising serum K⁺ by ~0.5 mEq/L. In states with upregulated/extrajunctional ACh receptors — burns, crush injury, denervation, prolonged immobility, muscular dystrophy, severe sepsis — the efflux is exaggerated and can be lethal.

BOARD TRAP

WRONG answer: using succinylcholine for rapid-sequence intubation in a patient with burns, crush injury, spinal cord injury/denervation, or prolonged immobilization.

Discriminator: in these patients use a non-depolarizing agent (e.g., rocuronium). The danger window for burns/denervation begins around 48–72 hours and persists for months.

9. Beta-blocker

WHAT YOU SEE

Beta-blocker blocking beta-2 uptake

WHAT IT ENCODES

Beta-2 adrenergic stimulation normally drives K⁺ into cells via the Na/K-ATPase, so non-selective beta-blockers (e.g., propranolol, labetalol) blunt cellular K⁺ uptake and cause a mild rise in serum K⁺ (usually < 0.5 mEq/L). The effect is usually clinically trivial alone but additive with other causes (CKD, ACEi).

BOARD TRAP

WRONG answer: blaming a cardioselective beta-1 blocker for significant hyperkalemia. Discriminator: it is the beta-2 blockade (non-selective agents) that matters; selective beta-1 agents have minimal effect. Conversely, the beta-2 AGONIST albuterol is a treatment that shifts K⁺ in.

10. Rhabdomyolysis

WHAT YOU SEE

Breaking bodies, high CK/PO₄/urate

WHAT IT ENCODES

Skeletal muscle breakdown releases intracellular contents into the blood: K⁺, phosphate, uric acid, myoglobin, and markedly elevated creatine kinase (CK often > 5,000–10,000, frequently > 15,000). The triad of high K⁺, high PO₄, and high CK with low calcium (calcium deposits in injured muscle) points here. Myoglobinuria can cause pigment-induced AKI, which then compounds the hyperkalemia.

BOARD TRAP

WRONG answer: forgetting that rhabdo causes HYPOcalcemia early. Discriminator: the released phosphate binds calcium and calcium deposits in damaged muscle, so early hypocalcemia is expected — do not over-correct it early, as calcium can mobilize back out during recovery and cause rebound hypercalcemia.

11. Tumor lysis

WHAT YOU SEE

Tumor lysis syndrome, high K/PO₄/LDH

WHAT IT ENCODES

Tumor lysis syndrome follows cytotoxic therapy (or spontaneously in bulky, high-turnover tumors like Burkitt lymphoma, ALL) and dumps intracellular contents: hyperkalemia, hyperphosphatemia, hyperuricemia, elevated LDH, and resultant HYPOcalcemia (phosphate binds calcium). It causes AKI from uric acid/calcium-phosphate crystals.

Prophylaxis is hydration plus allopurinol or rasburicase.

BOARD TRAP

WRONG answer: giving calcium for the hypocalcemia of TLS routinely. Discriminator: in hyperphosphatemic states, exogenous calcium promotes calcium-phosphate precipitation and worsens renal injury — treat hypocalcemia only if symptomatic, and prioritize lowering phosphate and uric acid.

12. Hemolysis/burns

WHAT YOU SEE

Massive transfusion, thermal injury

WHAT IT ENCODES

Massive intravascular hemolysis, large-volume transfusion of stored (potassium-leaking) red cells, and thermal injury all release intracellular K⁺ into the circulation. Note the separate, common artifact: PSEUDOhyperkalemia from in-vitro hemolysis of a poorly drawn/fist-clenched sample, thrombocytosis, or leukocytosis — a normal ECG with isolated high K⁺ should prompt a repeat draw.

BOARD TRAP

WRONG answer: treating an isolated high K⁺ from a hemolyzed specimen as true hyperkalemia. Discriminator: pseudohyperkalemia has no ECG changes and a 'hemolyzed' lab comment or extreme platelet/WBC counts; redraw without a tourniquet/fist-clenching before treating.

13. Familial HyperKPP

WHAT YOU SEE

SCN4A channelopathy, episodic weakness

WHAT IT ENCODES

Familial hyperkalemic periodic paralysis is an autosomal-dominant SCN4A skeletal muscle sodium-channel mutation causing episodic flaccid weakness, often triggered by rest after exercise, fasting, cold, or potassium ingestion; attacks last minutes to hours with mildly elevated K⁺. Contrast with hypokalemic periodic paralysis (CACNA1S/SCN4A), where attacks occur with LOW K⁺ after carbohydrate/insulin or exercise.

BOARD TRAP

WRONG answer: confusing hyper- and hypokalemic periodic paralysis. Discriminator: HyperKPP attacks are precipitated by potassium loads and rest-after-exercise and improve with carbohydrate/beta-agonist; hypoKPP attacks follow heavy carbohydrate meals and improve with potassium. Check the K⁺ DURING an attack.

14. Correct the trigger

WHAT YOU SEE

Pearl scroll: leak not excess

WHAT IT ENCODES

For redistribution hyperkalemia the cure is mechanism-directed: insulin (+/- glucose) for insulin deficiency/hyperosmolality, beta-2 agonist (albuterol) to drive K⁺ in, fluids for the dehydrated diabetic, and digoxin Fab for dig toxicity. Because total body K⁺ is normal, you generally do NOT need binders or dialysis once the trigger is reversed — and you must anticipate post-treatment hypokalemia.

BOARD TRAP

WRONG answer: managing a shift identically to a true K⁺ overload. Discriminator: the leak is positional. Treat the trigger and recheck K⁺ — if you also strip out total-body K⁺, the patient overshoots into hypokalemia when cells reclaim their K⁺.
